

Methodologies for Dependent Data Fusion & Their Equivalences Derived from Information Filtering

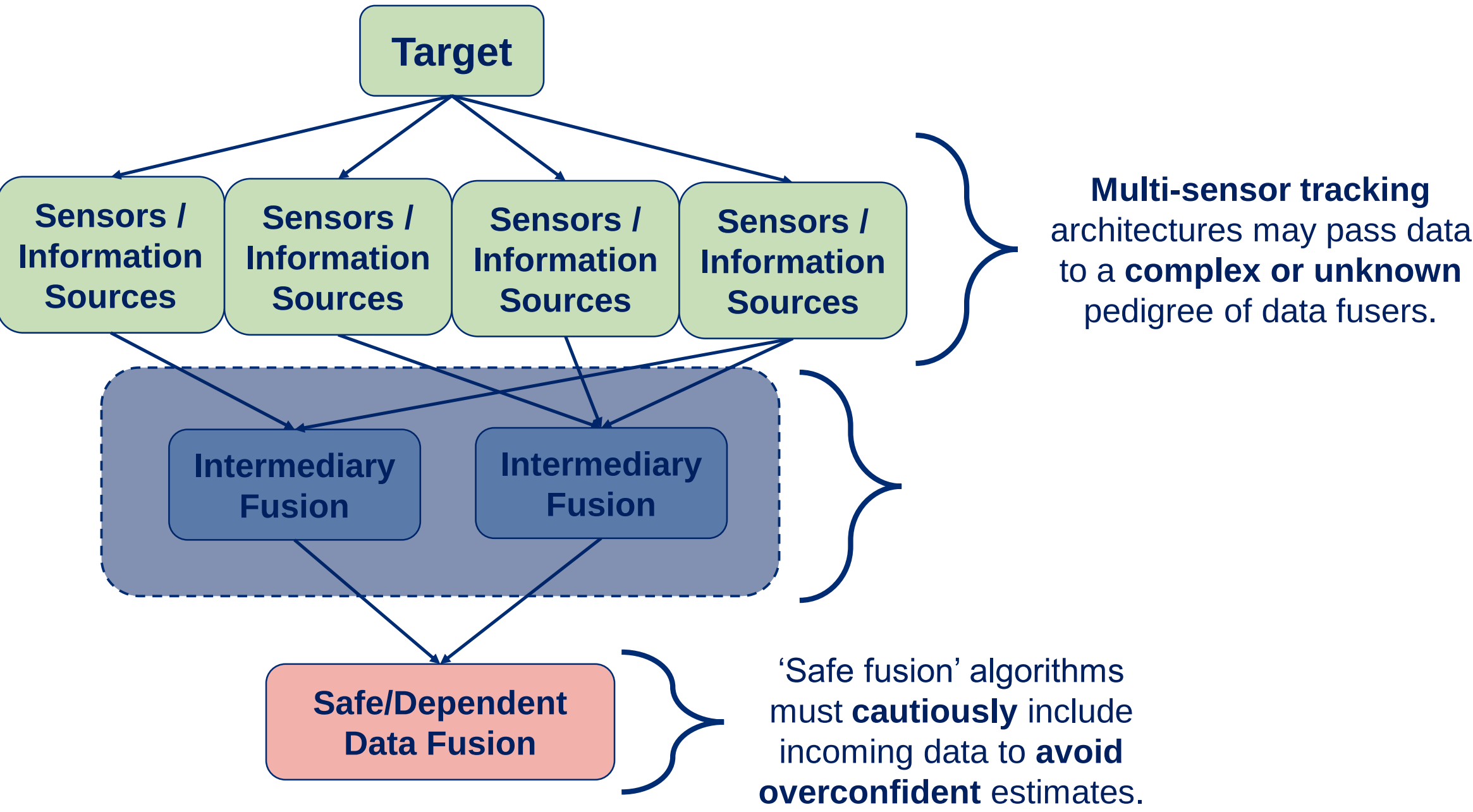
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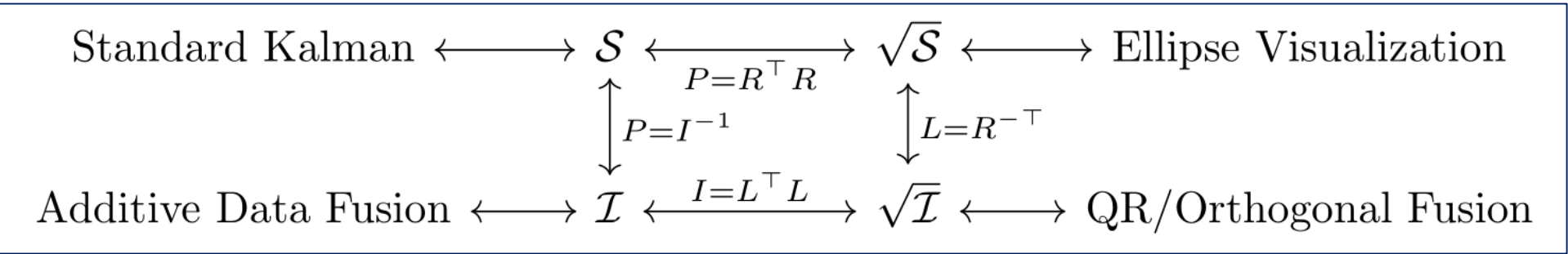


Principles & Premise:

The fusion of matrix information arising from dependent or decentralized sources is of key importance to multi-sensor covariance or square-root-information tracking. We explore the exclusive union of information as expressed through factorizations and orthogonal transformations of sets of data matrices.

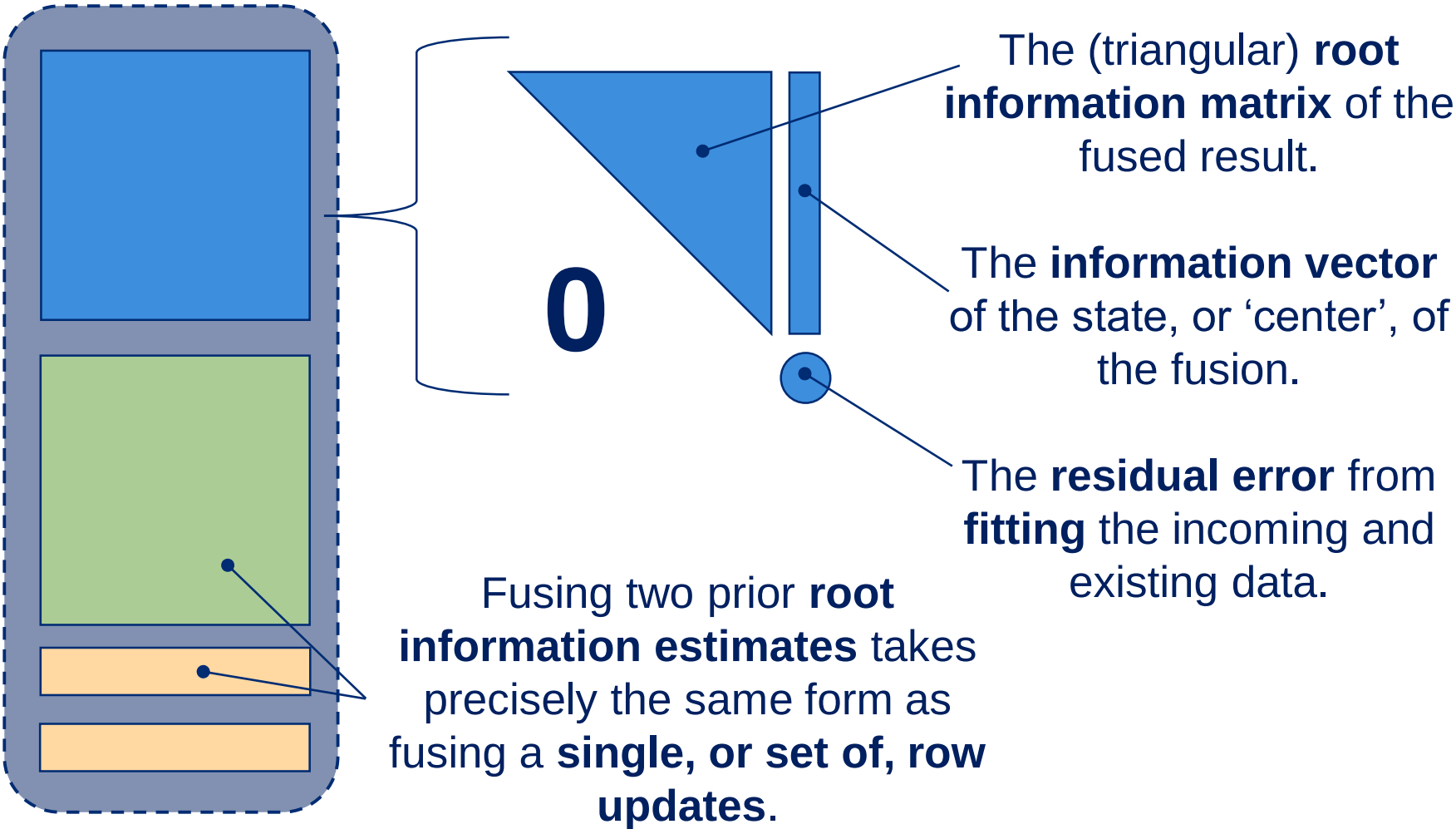


Premise of Methods: Determine 'how much' of the incoming data is shared with the known uncertainties and include only 'the difference'.



Anatomy of Root Fusion

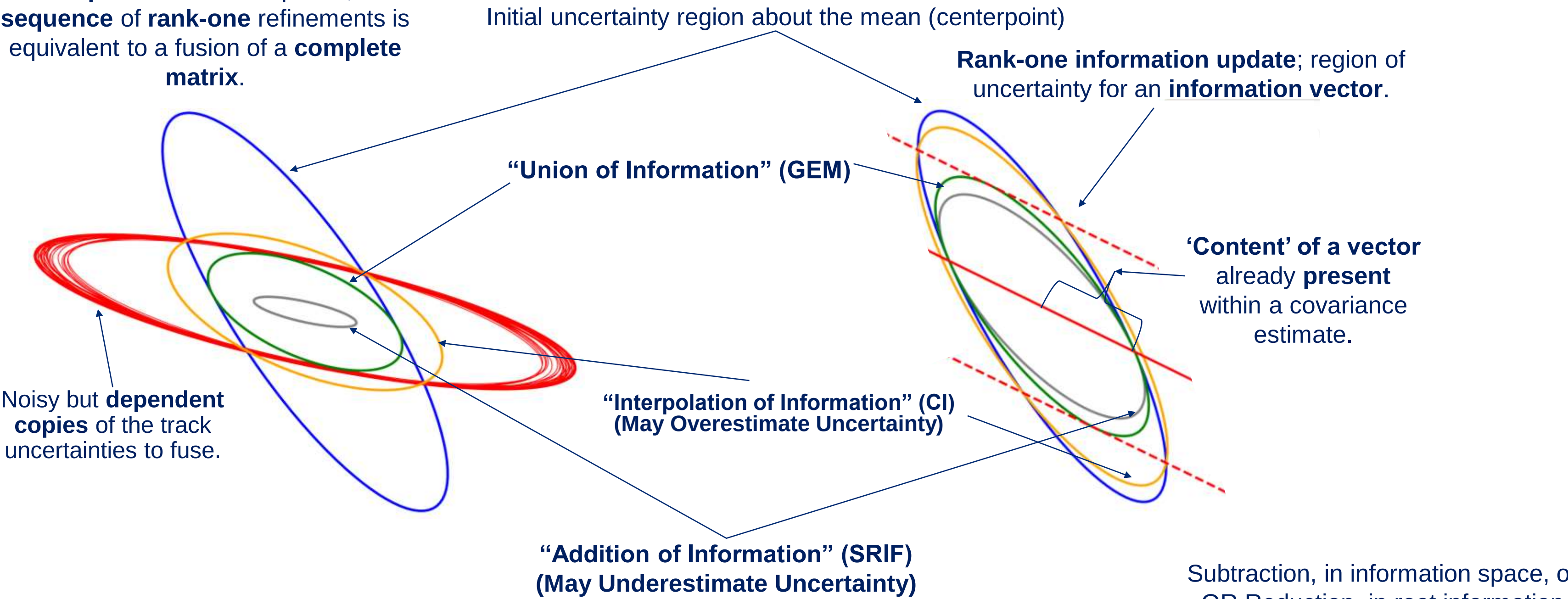
- Data fusion can occur in any such space under the axioms that **information is additive** and **root matrices are equivalent under left orthogonal transformations**.
- The Square Root Information Filter is an elegant and numerically efficient framework for updating root information matrices via the **QR decomposition** of the set of inputs.



- Fusion:** Track-to-track-fusion (T2TF) involves the combination of state and/or uncertainty estimates from distinct sources into a single, updated model. Tracks, or data history, are fused in accordance with their **certainty**.
 - By incorporating both the track state along with its covariance, a matrix of uncertainties between state variables, measurement-based T2TF algorithms such as, GEM and QR Reduction, theoretically serve to include weighted measurements in the decision process.
- GEM:** the Generalized Eigenvalue Method is a powerful framework which establishes **the intersection of the input information**, the quantity otherwise **double-counted** in fusion. Visually, GEM generates the largest ellipse **completely internal** to the inputs.
 - More precisely, the **GEM Basis**, found from **rotations** and the **Singular Value Decomposition**, provides a simultaneous basis of row vectors where comparisons of **singular values** permit a fusion algorithm to distinguish between **redundance and novelty**.

Under the appropriate **basis and/or independence** assumptions, a **sequence of rank-one** refinements is equivalent to a fusion of a **complete matrix**.

Visualizing Data Fusion – 2D



- Covariance, Information, Root Covariance, and Root Information** matrices are 'versions' of expressing uncertainty, akin to time/frequency domains. **Equivalences were sought and deduced** amongst the methods of data fusion in each space.
- Inclusion/Exclusion:** The explored data fusion algorithms involve a common basis of row vectors to compare 'new' and 'old' 'content' before omitting the doubly-included intersection.
- Future Research:** The authors are interested with the continued development of the **Safe-QR reduction algorithm**, the **tangency** of fused ellipses, and determining the **statistical interpretation**, via **likelihood functions**, of the these geometric methods.

Subtraction, in information space, or QR Reduction, in root information space, removes the intersection:

$$qr \left[\sqrt{X_A^2 - X_{A \cap B}^2}, \sqrt{X_B^2 - X_{A \cap B}^2}, X_A \cap X_B \right]$$

$$I_A + I_B - I_A \cap I_B$$

'Safe' single rank (i.e. one row) measurement updates serve to include **only the 'novel'** contributions of an update.

'Mutual Information Likelihood Optimization' (MILO) vectors correspond to **intersection points** and are **equally present** in both root matrices.

Project Roadmap

"Rotate and Compare": Each method employs **left orthogonal matrices** to align **rows in the same orientation** for comparison. These **rotations** do not leave the equivalence class of the covariance matrix.

Through the **Generalized Singular Value Decomposition**, a bridge between **row vector methods**, the **classical data fusion methods**, and the family of **GEM** fusions was rigorously established. In fact, the **GEM Basis** is precisely the GSVD singular vectors.

